

ATrax INJ
Cisatracurium Besylate Injection USP Solution for Injection or Infusion 2 mg/ml

Product Name

Atrax Inj Cisatracurium Besylate Injection USP Solution for Injection or Infusion 2 mg/ml

Name and Strength of Active Ingredient (s)

Cisatracurium besylate 2.676 mg equivalent to Cisatracurium 2mg.

Product Description

Atrax Inj Cisatracurium Besylate Injection USP Solution for Injection or Infusion 2 mg/ml is A clear, colorless to pale yellow or greenish colored solution.

Description of product after dilution: A clear colorless to pale yellow or greenish colored solution.

Pharmacodynamics / Pharmacokinetics Pharmacodynamic properties

Pharmacotherapeutic group Neuromuscular blocker

Mechanism of action and Pharmacodynamic effects:

Pharmacodynamic properties

Cisatracurium is an intermediate-duration, non-depolarising benzylisoquinolinium skeletal muscle relaxant.

Mode of action

Cisatracurium binds to cholinergic receptors on the motor end-plate to antagonise the action of acetylcholine, resulting in a competitive block of neuromuscular transmission. This action is readily reversed by anti-cholinesterase agents such as neostigmine or edrophonium.

The ED95 (dose required to produce 95% depression of the twitch response of the adductor pollicis muscle to stimulation of the ulnar nerve) of Cisatracurium is estimated to be 0.05mg/kg bodyweight during opioid anaesthesia (thiopentone /fentanyl midazolam).

The ED95 of Cisatracurium in children during halothane anaesthesia is 0.04 mg/kg.

Pharmacokinetic properties

Following administration of radiolabeled Cisatracurium, 95% of the dose was recovered in the urine; less than 10% of the dose was excreted as unchanged parent drug. Laudanosine, a metabolite of Cisatracurium (and Atracurium) has been noted to cause transient hypotension and, in higher doses, cerebral excitatory effects when administered to several animal species. The relationship between CNS excitation and laudanosine concentrations in humans has not been established. Because Cisatracurium is three times more potent than Atracurium.

Distribution

The volume of distribution of Cisatracurium is limited by its large molecular weight and high polarity. The V_{ss} was equal to 145 mL/kg in healthy 19 to 64 year-old surgical patients receiving opioid anesthesia. The V_{ss} was 21% larger in similar patients receiving inhalation anesthesia.

Protein Binding

The binding of Cisatracurium to plasma proteins has not been successfully studied due to its rapid degradation at physiologic pH. Inhibition of degradation requires nonphysiological conditions of temperature and pH which are associated with changes in protein binding.

Metabolism

The degradation of Cisatracurium is largely independent of liver metabolism. Results from *in vitro* experiments suggest that Cisatracurium undergoes Hofmann elimination (a pH and temperature-dependent chemical process) to form laudanosine. Laudanosine is further metabolized to desmethyol metabolites which are conjugated with glucuronic acid and excreted in the urine.

Elimination

Clearance and Half-life

Mean CL values for Cisatracurium ranged from 4.5 to 5.7 mL/min/kg in studies of healthy surgical patients. Compartmental pharmacokinetic modeling suggests that approximately 80% of the CL is accounted for by Hofmann elimination and the remaining 20% by renal and hepatic elimination. These findings are consistent with the low magnitude of interpatient variability in CL (16%) estimated as part of the population PK/PD analyses and with the recovery of parent and metabolites in urine. Following ^{14}C -Cisatracurium administration to 6 healthy male patients, 95% of the dose was recovered in the urine (mostly as conjugated metabolites) and 4% in the feces; less than 10% of the dose was excreted as unchanged parent drug in the urine. In 12 healthy surgical patients receiving non-radiolabeled Cisatracurium who had Foley catheters placed for surgical management, approximately 15% of the dose was excreted unchanged in the urine.

Indications

Cisatracurium Besylate Injection is an intermediate-duration, non-depolarising neuromuscular blocking agent for intravenous (i.v.) administration. Cisatracurium Besylate Injection is indicated for use during surgical and other procedures and in intensive care. It is used as an adjunct to general anaesthesia, or sedation in the Intensive Care Unit (ICU), to relax skeletal muscles, and to facilitate tracheal intubation and mechanical ventilation.

Cisatracurium Besylate Injection contains no antimicrobial preservative and is intended for single patient use.

Recommended Dose and Mode of administration

As with other neuromuscular blocking agents, monitoring of neuromuscular function is recommended during the use of Cisatracurium Besylate Injection in order to individualise dosage requirements.

● Use by I.V. bolus injection in adults

Tracheal Intubation

The recommended intubation dose of Cisatracurium Besylate Injection for adults is 0.15 mg/kg administered rapidly over 5 to 10 seconds. This dose produces good to excellent conditions for tracheal intubation 120 seconds following injection.

Higher doses will shorten the time to onset of neuromuscular block. Table 1 summarises mean pharmacodynamic data when Cisatracurium Besylate Injection was administered at doses of 0.1 to 0.4 mg/kg to healthy adult patients during opioid (thiopentone/fentanyl/midazolam) or propofol anaesthesia.

Table 1: Mean Pharmacodynamic Data Following a Range of Cisatracurium Besylate Injection Doses

Initial Cisatracurium Besylate Injection dose (mg/kg)	Anaesthetic background	Time to 90% T ₁ ^a suppression (minutes)	Time to maximum T ₁ ^a suppression (minutes)	Time to 25% spontaneous T ₁ ^a recovery (minutes)
0.1	Opioid	3.4	4.8	45
0.15	Propofol	2.6	3.5	55
0.2	Opioid	2.4	2.9	65
0.4	Opioid	1.5	1.9	91

^a Single twitch response as well as the first component of the Train-of-Four response of the adductor pollicis muscle following supramaximal electrical stimulation of the ulnar nerve.

Enflurane or isoflurane anaesthesia may extend the clinically effective duration of an initial dose of Cisatracurium Besylate Injection by as much as 15%.

Maintenance

Neuromuscular block can be extended with maintenance doses of Cisatracurium Besylate Injection. A dose of 0.03 mg/kg provides approximately 20 minutes of additional clinically effective neuromuscular block during opioid or propofol anaesthesia. Consecutive maintenance doses do not result in progressive prolongation of effect.

Spontaneous recovery

Once spontaneous recovery from neuromuscular block is underway, the rate is independent of the Cisatracurium Besylate Injection dose administered. During opioid or propofol anaesthesia, the median times from 25 to 75% and from 5 to 95% recovery are approximately 13 and 30 minutes, respectively.

Reversal

Neuromuscular block following Cisatracurium Besylate Injection administration is readily reversible with standard doses of anticholinesterase agents. The mean times from 25 to 75% recovery and to full clinical recovery (T₄:T₁ ratio more than or equal to 0.7) are approximately 2 and 5 minutes, respectively, following administration of the reversal agent at an average of 13% T₁ recovery.

● Use by I.V. bolus injection in children (aged 2 to 12 years of age)

Tracheal intubation

The recommended initial intubation dose of Cisatracurium Besylate Injection in children aged 2 to 12 years is 0.1 mg/kg administered over 5 to 10 seconds. The following table summarises mean pharmacodynamic data obtained during opioid anaesthesia. A dose of 0.1 mg/kg has a faster onset time, a shorter clinical effective duration and a faster spontaneous recovery profile than those observed in adults under similar anaesthesia conditions.

Initial Cisatracurium Besylate Injection dose (mg/kg)	Anaesthetic background	Time to 90% T ₁ suppression (minutes)	Time to maximum T ₁ suppression (minutes)	Time to 25% spontaneous T ₁ recovery (minutes)
0.1	Opioid	1.7	2.8	28
0.08	Halothane	1.7	2.5	31

Halothane may be expected to extend the clinical effective duration of Cisatracurium Besylate Injection by up to 20%. No information is available on the use of Cisatracurium Besylate Injection in children during isoflurane or enflurane anaesthesia but these agents may also be expected to extend the clinically effective duration of a dose of Cisatracurium Besylate Injection by up to 20%.

Although tracheal intubation has not been specifically studied in this age group, onset is faster than in adults and therefore intubation should also be possible within two minutes of administration.

Maintenance

Neuromuscular block can be extended with maintenance doses of Cisatracurium Besylate Injection injection. A dose of 0.02 mg/kg provides approximately 9 minutes of additional clinically effective neuromuscular block during halothane anaesthesia. Consecutive maintenance doses do not result in progressive prolongation of effect.

Spontaneous Recovery

Once recovery from neuromuscular block is underway, the rate is independent of the Cisatracurium Besylate Injection dose administered. During opioid or halothane anesthesia, the median times from 25 to 75% and from 5 to 95% recovery are approximately 11 and 28 minutes, respectively.

Reversal

Neuromuscular block following Cisatracurium Besylate Injection administration is readily reversible with standard doses of anticholinesterase agents. The mean times from 25 to 75% recovery and to full clinical recovery (T₄:T₁ ratio more than or equal to 0.7) are approximately 2 and 5 minutes, respectively, following administration of the reversal agent at an average of 13% T₁ recovery.

- **Use by I.V. infusion in adults and children (aged 2 to 12 years)**

Maintenance of neuromuscular block may be achieved by infusion of Cisatracurium Besylate Injection. An initial infusion rate of 3 micrograms/kg/min (0.18 mg/kg/h) is recommended to restore 89 to 99% T₁ suppression following evidence of spontaneous recovery. After an initial period of stabilisation of neuromuscular block, a rate of 1 to 2 micrograms/kg/min (0.06 to 0.12 mg/kg/h) should be adequate to maintain block in this range in most patients.

Reduction of the infusion rate by up to 40% may be required when Cisatracurium Besylate Injection is administered during isoflurane or enflurane anaesthesia (see *Interactions*).

The infusion rate will depend upon the concentration of Cisatracurium Besylate Injection in the infusion solution, the desired degree of neuromuscular block, and the patient's weight. Table 2 provides guidelines for delivery of undiluted Cisatracurium Besylate Injection.

Table 2: Infusion Delivery Rate of Cisatracurium Besylate Injection 2 mg/mL

Patient weight (kg)	Dose (micrograms/kg/min)				Infusion rate
	1.0	1.5	2.0	3.0	
20	0.6	0.9	1.2	1.8	mL/h
70	2.1	3.2	4.2	6.3	mL/h
100	3.0	4.5	6.0	9.0	mL/h

Steady rate continuous infusion of Cisatracurium Besylate Injection is not associated with a progressive increase or decrease in neuromuscular blocking effect.

Following discontinuation of infusion of Cisatracurium Besylate Injection, spontaneous recovery from neuromuscular block proceeds at a rate comparable to that following administration of a single bolus.

- **Neonates aged less than 2 years**

No dosage recommendation for paediatric patients under 2 years of age can be made until further information becomes available.

- **Elderly**

No dosing alterations are required in elderly patients. In these patients Cisatracurium Besylate Injection has a similar pharmacodynamic profile to that observed in young adult patients but, as with other neuromuscular blocking agents, it may have a slightly slower onset.

- **Patients with renal impairment**

No dosing alterations are required in patients with renal failure. In these patients Cisatracurium Besylate Injection has a similar pharmacodynamic profile to that observed in patients with normal renal function but it may have a slightly slower onset.

- **Patients with hepatic impairment**

No dosing alterations are required in patients with end-stage liver disease. In these patients Cisatracurium Besylate Injection has a similar pharmacodynamic profile to that observed in patients with normal hepatic function but it may have a slightly faster onset.

- **Patients with cardiovascular disease**

Cisatracurium Besylate Injection has been used effectively to provide neuromuscular block in patients undergoing cardiac surgery. When administered by rapid bolus injection (over 5 to 10 seconds) to patients with serious cardiovascular disease Cisatracurium Besylate Injection has not been associated with clinically significant cardiovascular effects at any dose studied (up to and including 0.4 mg/kg (8 x ED₉₅)).

- **ICU patients**

Cisatracurium Besylate Injection may be administered by bolus dose and/or infusion to adult patients in the ICU.

An initial infusion rate of Cisatracurium Besylate Injection of 3 micrograms/kg/min (0.18 mg/kg/h) is recommended for adult ICU patients. There may be wide inter-patient variation in dosage requirements and these may increase or decrease with time. In clinical studies the average infusion rate was 3 micrograms/kg/min [range 0.5 to 10.2 micrograms/kg/min (0.03 to 0.6 mg/kg/h)]. Table 3 provides guidelines for delivery of undiluted Cisatracurium Besylate Injection.

The median time to full spontaneous recovery following long-term (up to 6 days) infusion of Cisatracurium Besylate Injection in ICU patients was approximately 50 minutes.

Table 3: Infusion Delivery Rate of Cisatracurium Besylate Injection 5 mg/mL

Patient weight (kg)	Dose (micrograms/kg/min)				Infusion rate
	1.0	1.5	2.0	3.0	
70	0.8	1.2	1.7	2.5	mL/h
100	1.2	1.8	2.4	3.6	mL/h

The recovery profile after infusions of Cisatracurium Besylate Injection to ICU patients is independent of duration of infusion.

- **Patients undergoing hypothermic cardiac surgery**

There have been no studies of Cisatracurium Besylate Injection in patients undergoing surgery with induced hypothermia (25°C to 28°C). As with other neuromuscular blocking agents, the rate of infusion required to maintain adequate surgical relaxation under these conditions may be expected to be significantly reduced.

Contraindications

Cisatracurium Besylate Injection is contraindicated in patients known to have an allergic hypersensitivity to Cisatracurium Besylate or other bis-benzylisoquinolinium agents.

Warnings and Precautions

Cisatracurium Besylate Injection paralyses the respiratory muscles as well as other skeletal muscles but has no known effect on consciousness or pain threshold. Cisatracurium Besylate Injection should be only administered by or under the supervision of anaesthetists or other clinicians who are familiar with the use and action of neuromuscular blocking agents. Facilities for tracheal intubation, and maintenance of pulmonary ventilation and adequate arterial oxygenation should be available.

Caution should be exercised when administering Cisatracurium Besylate Injection to patients who have shown hypersensitivity to other neuromuscular blocking agents since a high rate of cross-sensitivity (greater than 50%) between neuromuscular blocking agents has been reported.

Cisatracurium Besylate Injection does not have significant vagolytic or ganglion- blocking properties. Consequently, Cisatracurium Besylate Injection has no clinically significant effect on heart rate and will not counteract the bradycardia produced by many anesthetic agents or by vagal stimulation during surgery.

Patients with myasthenia gravis and other forms of neuromuscular disease have shown greatly increased sensitivity to non-depolarising blocking agents. An initial dose of not more than 0.02 mg/kg Cisatracurium Besylate Injection is recommended in these patients.

Severe acid-base and/or serum electrolyte abnormalities may increase or decrease the sensitivity of patients to neuromuscular blocking agents.

Cisatracurium Besylate Injection has not been studied in patients with a history of malignant hyperthermia. Studies in malignant hyperthermia- susceptible pigs indicated that Cisatracurium Besylate Injection does not trigger this syndrome.

Cisatracurium Besylate Injection has not studied in patients with burns; however, as with other non- depolarising neuromuscular blocking agents, the possibility of increased dosing requirements and shortened duration of action must be considered if Cisatracurium Besylate injection is administered to these patients.

Cisatracurium Besylate Injection is hypotonic and must not be administered into the infusion line of a blood transfusion.

- **ICU Patients**

When administered to laboratory animals in high doses, laudanosine, a metabolite of Cisatracurium and atracurium, has been associated with transient hypotension and in some species, cerebral excitatory effects.

Consistent with the decreased infusion rate requirements of Cisatracurium Besylate Injection, plasma laudanosine concentrations are approximately one third those following atracurium infusion.

There have been rare reports of seizures in ICU patients who have received atracurium and other agents. These patients usually had one or more medical conditions predisposing to seizures (e.g. cranial trauma, hypoxic encephalopathy, cerebral oedema, viral encephalitis, uremia).

A causal relationship to laudanosine has not been established.

Interactions

Many drugs have been shown to influence the magnitude and/or duration of action of non- depolarising neuromuscular blocking agents, including the following:

Increased effect

Anaesthetics:

- volatile agents such as enflurane, isoflurane and halothane.
- ketamine.
- other non-depolarising neuromuscular blocking agents.

Other drugs:

- antibiotics: including the aminoglycosides, polymyxins, spectinomycin, tetracyclines, lincomycin and clindamycin.
- anti-arrhythmic drugs: including propranolol, calcium channel blockers, lignocaine, procainamide and quinidine.
- diuretics: including furosemide and possibly thiazides, mannitol and acetazolamide.

- magnesium salts.
- lithium salts.
- ganglion blocking drugs: trimetaphan, hexamethonium.

Rarely, certain drugs may aggravate or unmask latent myasthenia gravis or actually induce a myasthenic syndrome; increased sensitivity to non-depolarising neuromuscular blocking agents might result. Such drugs include various antibiotics, beta-blockers (propranolol, oxprenolol), anti-arrhythmic drugs (procainamide, quinidine), anti-rheumatic drugs (chloroquine, D- penicillamine), trimetaphan, chlorpromazine, steroids, phenytoin and lithium.

Administration of suxamethonium to prolong the effects of non-depolarising neuromuscular blocking agents may result in a prolonged and complex block which can be difficult to reverse with anticholinesterases.

Decreased effect

- prior chronic administration of phenytoin or carbamazepine.
- treatment with anticholinesterases, commonly used in the treatment of Alzheimer's disease e.g. donepezil, may shorten the duration and diminish the magnitude of neuromuscular blockade with cisatracurium.

No effect

- prior administration of suxamethonium has no effect on the duration of neuromuscular block following bolus doses of Cisatracurium Besylate Injection or on infusion rate requirements.

Pregnancy and Lactation

There are no adequate data from the use of Cisatracurium Besylate Injection in pregnant women. Animal studies are insufficient with respect to effects on pregnancy, embryonal/foetal development, parturition and postnatal development. The potential risk for humans is unknown.

Cisatracurium Besylate Injection should not be used during pregnancy.

It is not known whether Cisatracurium or its metabolites are excreted in human milk.

Undesirable Effects

Data from pooled internal clinical trials were used to determine the frequency of very common to uncommon adverse reactions. The following convention has been used for the classification of frequency:- very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1000$ to $< 1/100$), rare ($\geq 1/10,000$ to $< 1/1000$), very rare ($< 1/10,000$).

Clinical Trial Data

Cardiac disorders
Common Bradycardia

Vascular disorders
Common Hypotension
Uncommon Cutaneous flushing

Respiratory, thoracic and mediastinal disorders
Uncommon Bronchospasm

Skin and subcutaneous tissue disorders
Uncommon Rash

Post-Marketing Data

Immune system disorders
Very rare Anaphylactic reaction

Anaphylactic reactions of varying degrees of severity have been observed after the administration of neuromuscular blocking agents. Very rarely, severe anaphylactic reactions have been reported in patients receiving Cisatracurium Besylate Injection. in conjunction with one or more anaesthetic agents.

Musculoskeletal and connective tissue disorders
Very rare Myopathy, muscle weakness

There have been some reports of muscle weakness and/or myopathy following prolonged use of muscle relaxants in severely ill patients in the ICU. Most patients were receiving concomitant corticosteroids. These events have been reported infrequently in association with Cisatracurium Besylate Injection and a causal relationship has not been established.

Overdose and treatment

Prolonged muscle paralysis and its consequences are expected to be the main signs of over dosage with Cisatracurium Besylate Injection.

Management

It is essential to maintain pulmonary ventilation and arterial oxygenation until adequate spontaneous respiration returns. Full sedation will be required since consciousness is not impaired by Cisatracurium Besylate Injection. Recovery may be accelerated by the administration of anti-cholinesterase agents once evidence of spontaneous recovery is present.

Instructions for Use

Incompatibilities
Cisatracurium Besylate Injection is not chemically stable when diluted with Lactated Ringer's Injection. Since Cisatracurium Besylate Injection is stable only in acidic solutions, it should not be mixed in the same syringe or administered simultaneously through the same needle with alkaline solutions, e.g. sodium thiopentone. It is not compatible with ketorolac trometamol or propofol injectable emulsion.

Special precautions for disposal and other handling

Any unused product or waste material should be disposed of in accordance with local requirements.
Diluted to a concentration of 0.1 mg cisatracurium/ml is physically and chemically stable for 24 hours at 2-8°C in sodium chloride 9 mg/ml (0.9%) solution and in glucose 50 mg/ml (5%) solution.
Cisatracurium has been shown to be compatible with the following commonly used peri-operative medicinal products, when mixed in conditions simulating administration into a running intravenous infusion via a Y-site injection port: alfentanil hydrochloride, droperidol, fentanyl citrate, midazolam hydrochloride and sufentanil citrate.
Where other agents are administered through the same indwelling needle or cannula as Cisatracurium-hameln, it is recommended that each medicinal product be flushed through with an adequate volume of a suitable intravenous fluid, e.g., sodium chloride 9 mg/ml (0.9%) solution.

Storage Condition

Store at 2 to 8 °C. Protect from light. Do not freeze.

Dilution: Upon dilution with proposed diluents (0.9% Sodium Chloride Injection, 5% Dextrose Injection) cisatracurium besylate Injection is compatible and stable upto 24 hours when stored at labelled storage condition 2 - 8°C. From a microbiological point of view, it should be used immediately unless dilution has taken place in controlled and validated aseptic conditions.

Shelf Life

Unopened vial: 21 months

Shelf life after first opening:

The product should be used immediately after opening the vial

Prepared infusion solution:

Chemical and physical in-use stability has been demonstrated for 24 hours at 2-8°C

From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2-8°C

Dosage Forms and Packaging available

A clear, colorless to pale yellow or greenish colored solution filled in 5 mL USP type I clear glass vial, stopper with 20 mm ready to sterilize Bromobutyl rubber stopper and sealed with 20 mm aluminum flip-off seals. Such labeled vials is placed in a mono carton with pack insert and such 10 vials is packed in outer carton.

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